

SAMPLING DISTRIBUTIONS & THE CLT

new 2.12, 3.1–3.2, 4.1 • old 5.1–5.7

ZONE A · THE FORMULAS + CONDITIONS

1 · RANDOM: START WITH A RANDOM SAMPLE

SAMPLE PROPORTION $\hat{p}$ Population proportion p	SAMPLE MEAN $\bar{x}$ Population mean μ , SD σ
$\mu_{\hat{p}} = p$	$\mu_{\bar{x}} = \mu$
$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$	$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

2 · 10%: $n < 0.10 \cdot N$ when sampling without replacement.
Draws are close enough to independent to use these SD formulas.

3 · LARGE COUNTS

The distribution of $\hat{p}$ is approximately normal when
 $np \geq 10$ and $n(1-p) \geq 10$.

3 · NORMAL / CENTRAL LIMIT THEOREM

Normal population $\Rightarrow$ normal $\bar{x}$, for any n .
Otherwise, CLT gives an **approximate** normal shape
for large n . The $n \geq 30$ guideline is approximate;
strong skew can need a larger sample.

ZONE B · PARAMETERS VS STATISTICS

	Parameter	Statistic	
Mean	μ	$\bar{x}$	Parameters are Greek or unhatted. Statistics wear a hat or a bar (or use the sample symbols s and b). Unbiased: the sampling distribution is centered at the parameter. Variability shrinks with $\sqrt{n}$: quadrupling n halves the SD.
SD	σ	s	
Proportion	p	$\hat{p}$	
Slope	β	b	

ZONE C · SAY IT IN WORDS

“The sampling distribution of _____ is the distribution of that statistic from all possible samples of size $n = \rule{1cm}{0.4pt}$ from _____.”
One sample gives one statistic; repeated samples give a distribution.

ZONE D · WATCH OUT

KEEP n AND N SEPARATE

n is sample size; N is population size.
Sample size controls these SDs.
Use N for the 10% check.

SD VS ESTIMATED SD

$\sigma_{\bar{x}} = \sigma / \sqrt{n}$ needs population SD σ . With sample SD s ,
 $s / \sqrt{n}$ is an **SE** estimating the SD of $\bar{x}$. See P13.